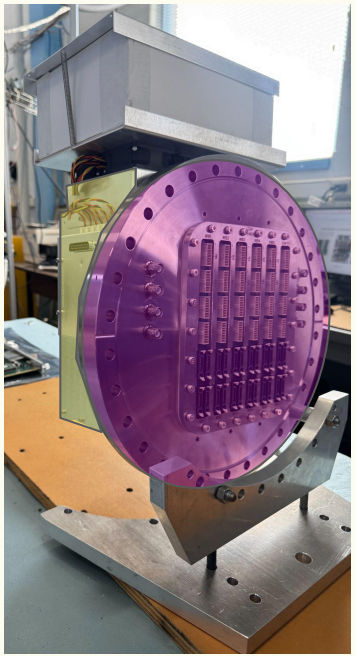
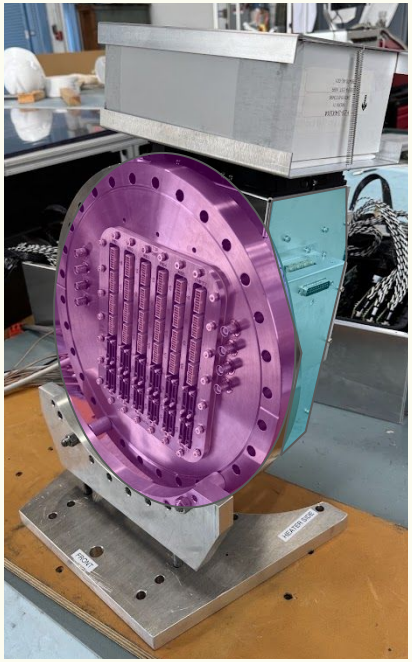
# WIEC QC Test Stand



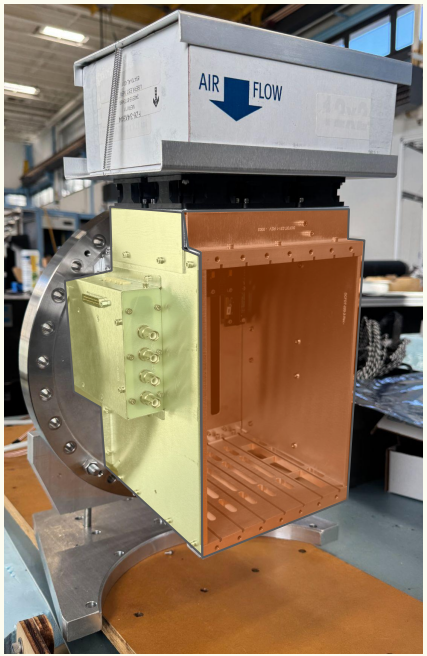
Fan Side | Front



Front | Heater Side




Rear | Heater Side

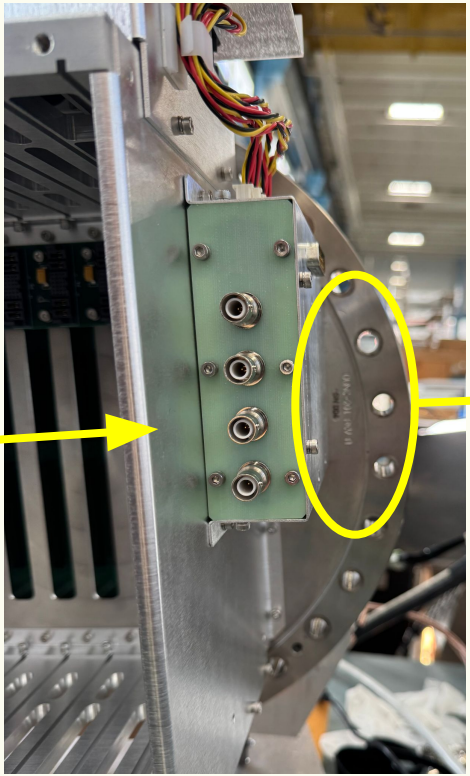



Fan Side | Rear

# Find WIEC Serial - On Fan Side




Rear



Fan Side




This serial is:

DUNE-201 REV A SN 004

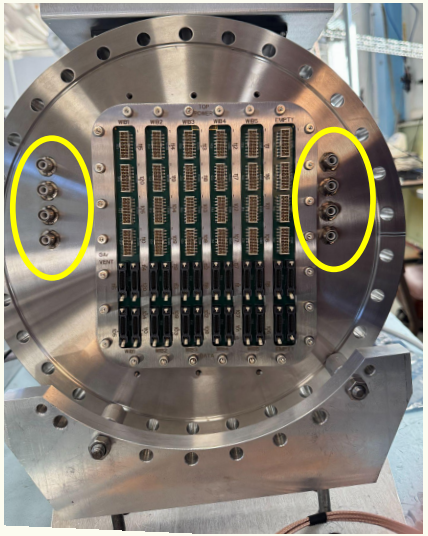
Make sure to input exactly what is on your WIEC.

# Visual Inspection

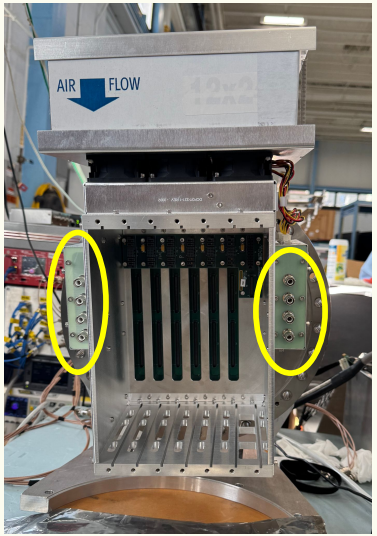
## SHV Connectors

Check every connector for any deviations from their expected shapes.

| **PASS**<br>Good SHV Connectors | **FAIL**<br>Damaged SHV Connectors |
| --- | --- |
|   |   |



Front x8



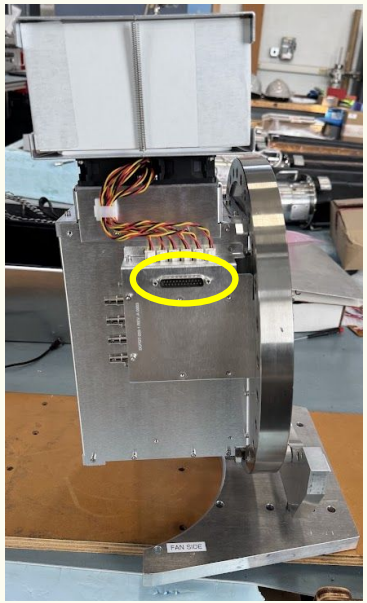

Rear x8

# Visual Inspection

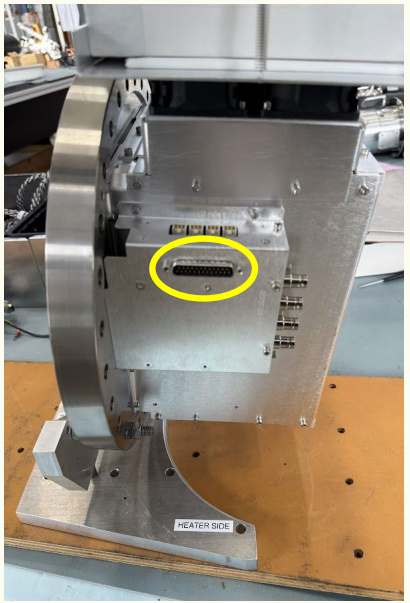
## Fan and Heater Connectors

Check every connector for any deviations from their expected shapes and that screws are in place.

| | |
|---|---|
| **PASS**<br><br>Good Shape<br>Screws circled |  |
| **FAIL**<br><br>Missing Screws |  |




Fan side x1

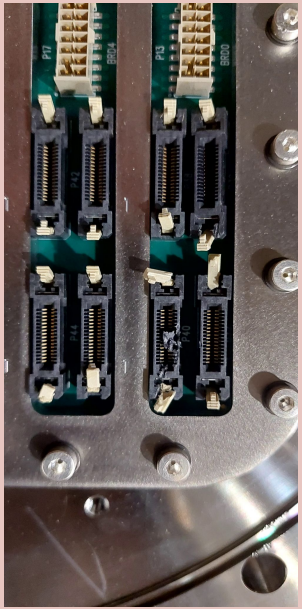


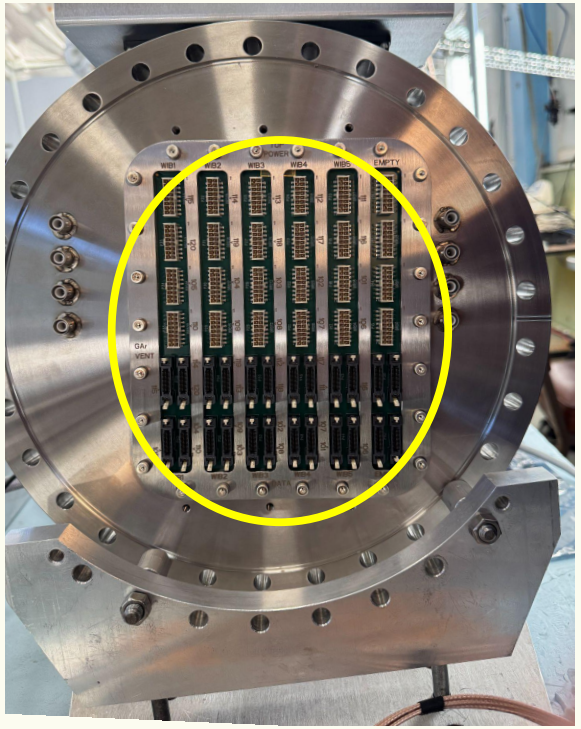

Heater side x1

# Visual Inspection

## Flange Board

Check every connector for any deviations from their expected shapes.

| PASS | FAIL |
| --- | --- |
| Flange Board Connectors | Damaged Connectors |
|  |  |



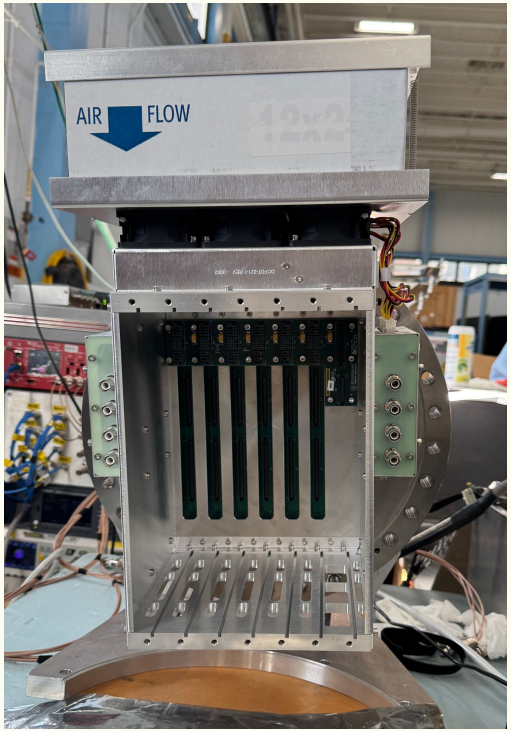
Front Flange Board

# Visual Inspection

## Air Filter

Check the air filter is installed in the correct direction.

| | |
|---|---|
| **PASS**<br><br>Air Filter Arrow Correct |  |
| **FAIL**<br><br>Air Filter Arrow Incorrect |  |

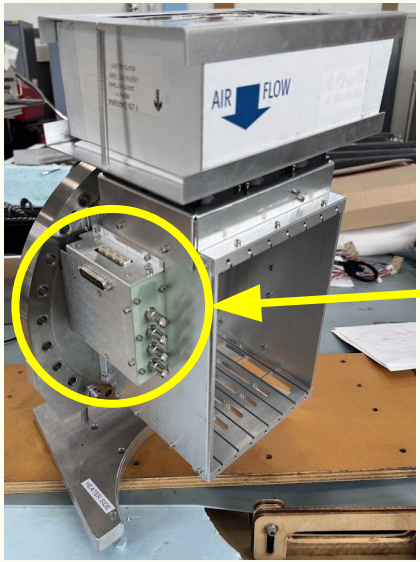


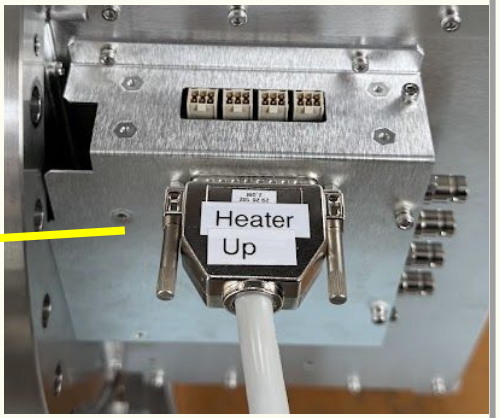

Rear

# Setup

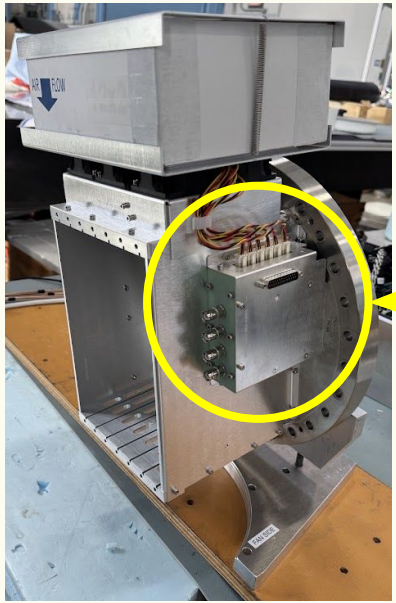
## Heater and Fan Connectors

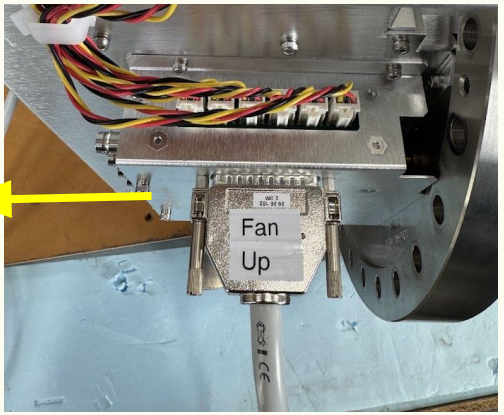
- Heater on heater side, Fan on fan side.
- Screw in the screws on the connector.




Heater Side



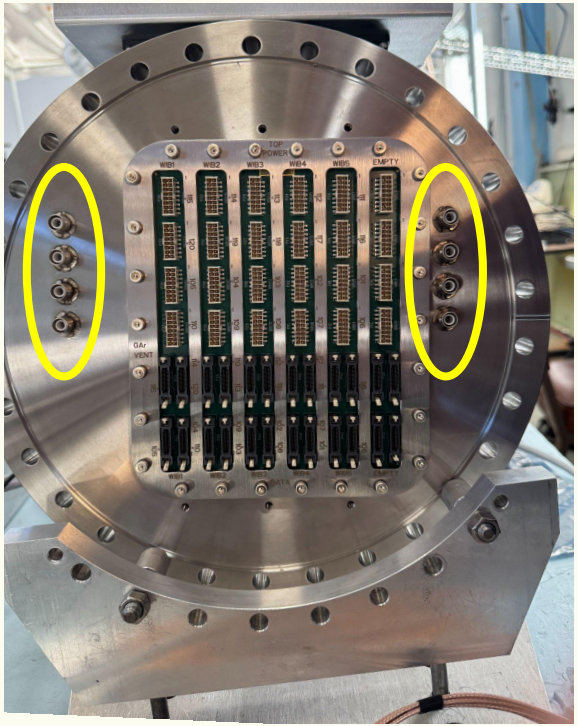



Fan Side

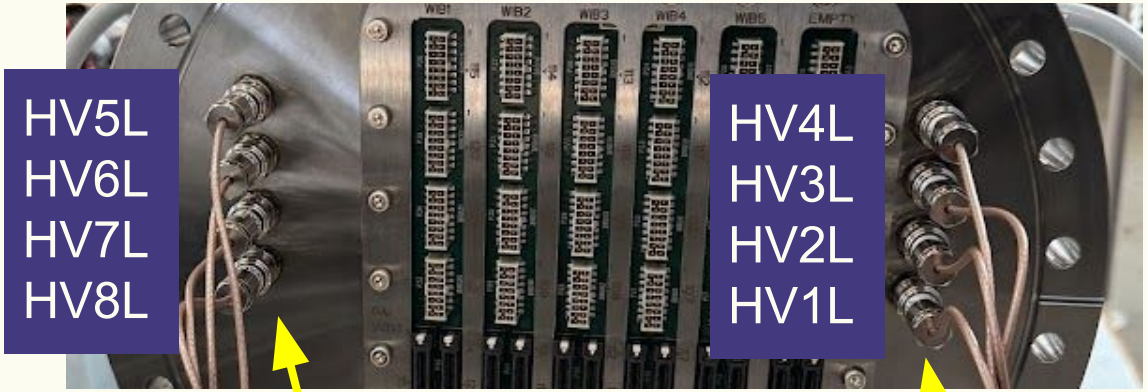
# Setup

## Front SHV

- Plug in SHV cables ending in **L** to their designated connectors.
- Turn connectors to ensure they are fully seated.



Front





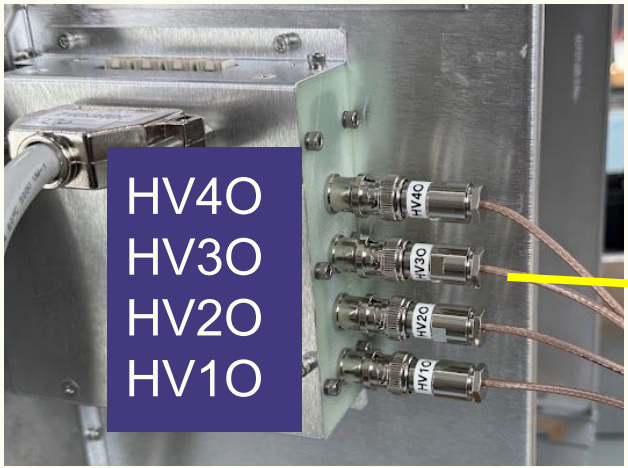




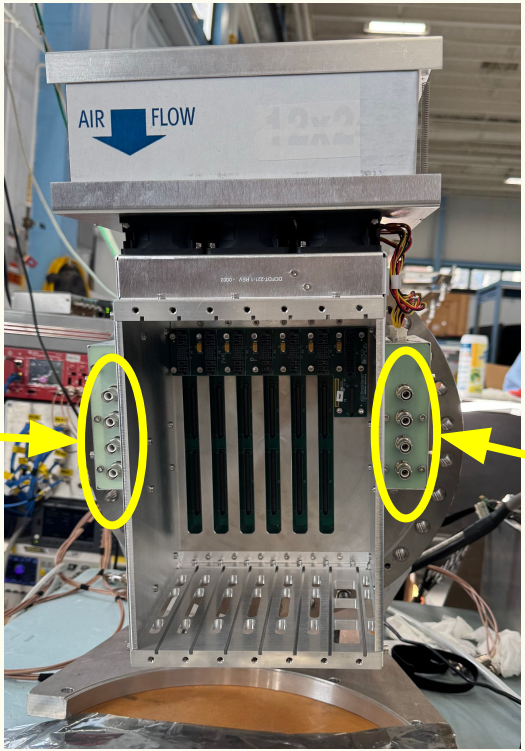
# Setup

## Rear SHV

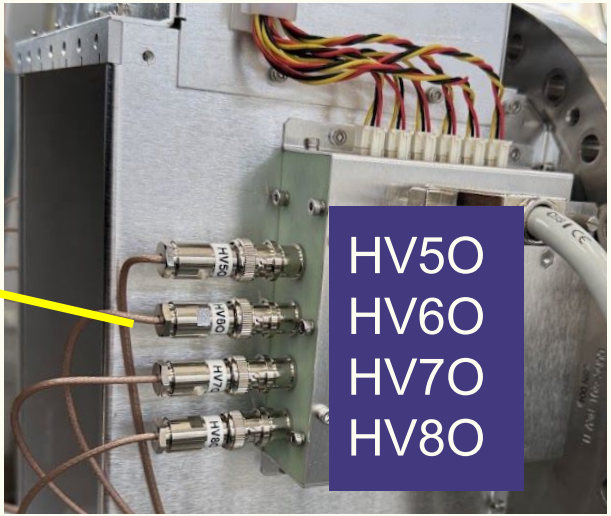
- Plug in SHV cables ending in **O** to their designated connectors.
- Turn connectors to ensure they are fully seated.




Heater Side




Rear

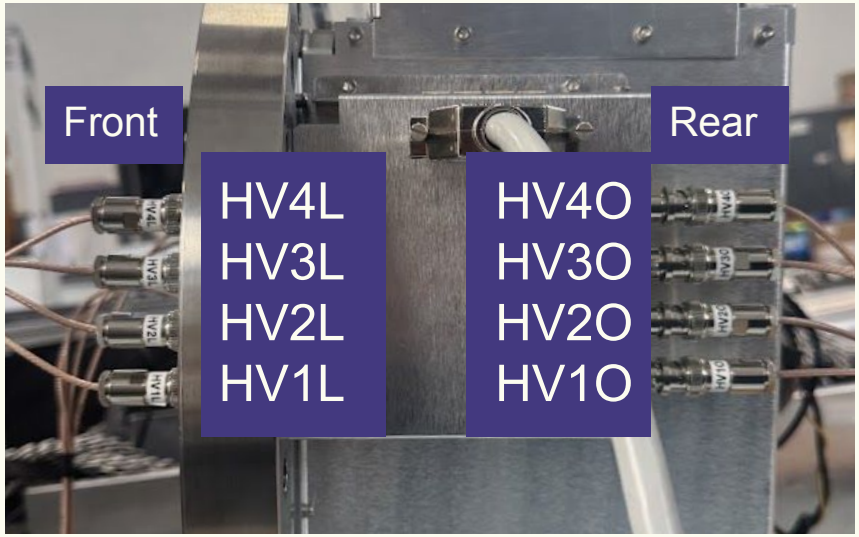



Fan Side

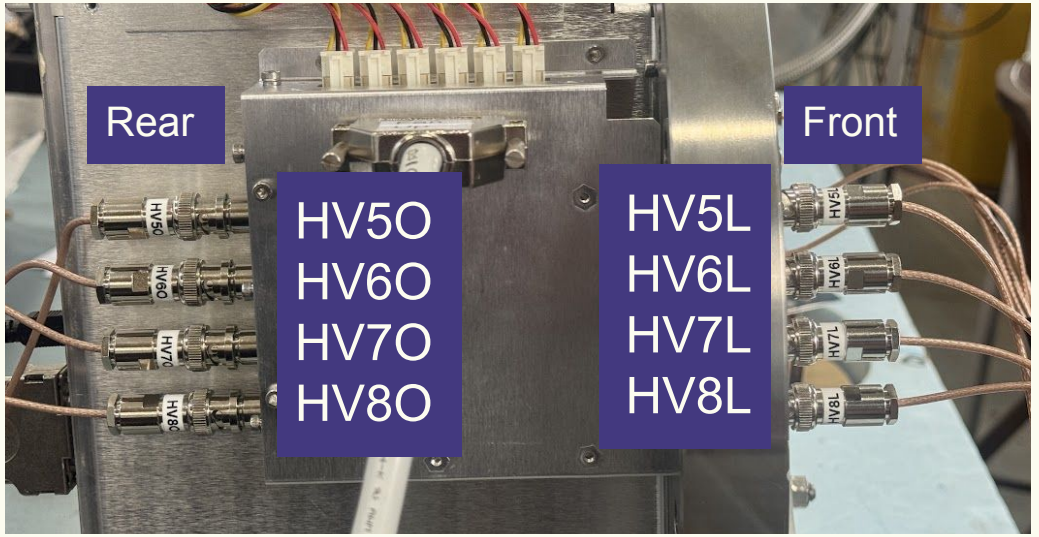
# Setup

## SHV Check

- If plugged in wrong, the WIEC will be permanently damaged.
- **L** cables on the front, **O** cables on the rear.




Heater Side



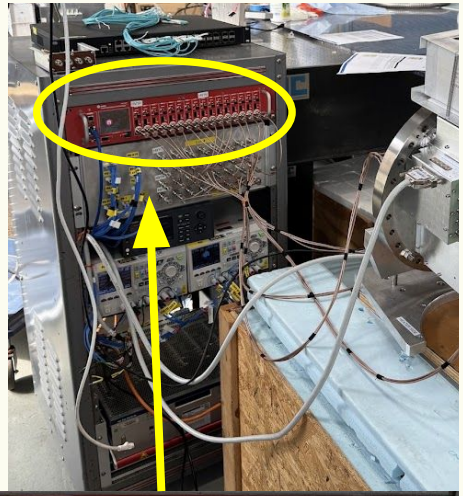

Fan Side

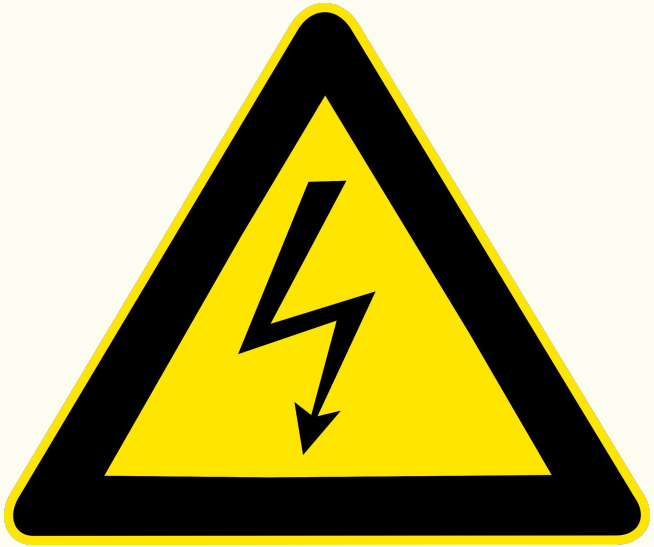
# Setup

## High Voltage Warning

**SEVERE ELECTRIC SHOCK RISK!**

- Do not touch any part of the test hardware when the high voltage supply lights are on.
- When on, the lights will be illuminated in red on half of the power supply.
- Make sure to check both sides before touching the WIEC.







# WIB/FEMB Test

## Install PTC



PTC



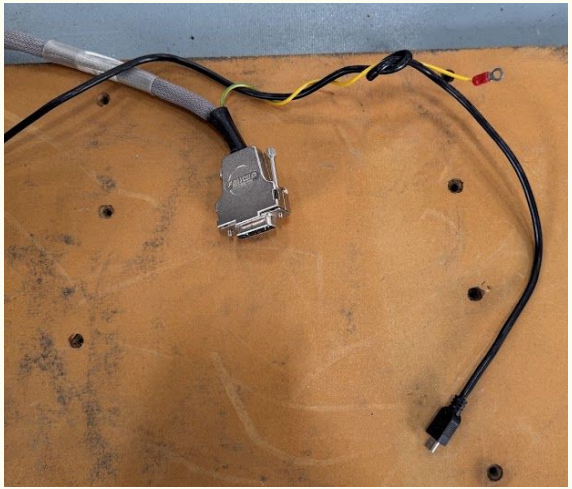

Slide the PTC into the slot closest to the fan side until you can screw in the top and bottom screws, circled.

- Screw in the screws until it becomes difficult to screw them further.
- Press the board further into the slot.
- Alternate screwing in and pressing the PTC until the board is flush with the crate.

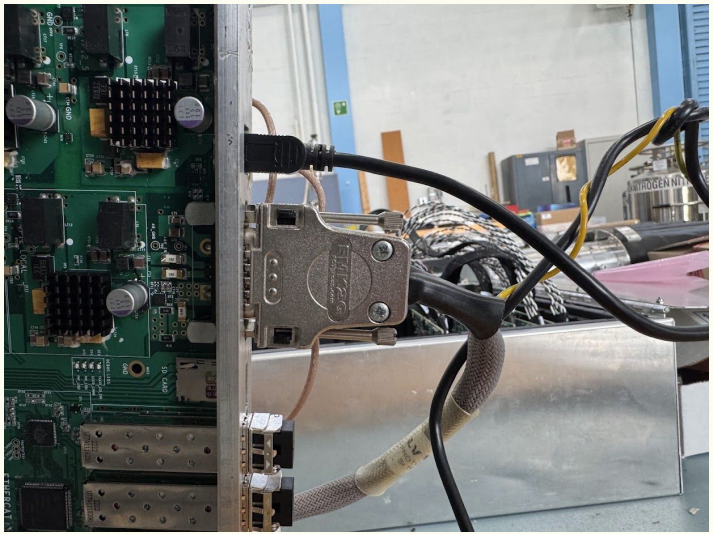


# WIB/FEMB Test

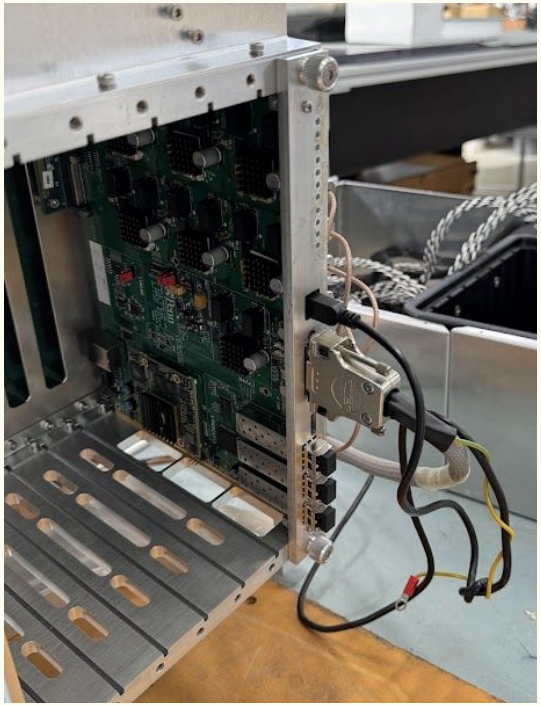
## Attach PTC Cables



PTC Cables



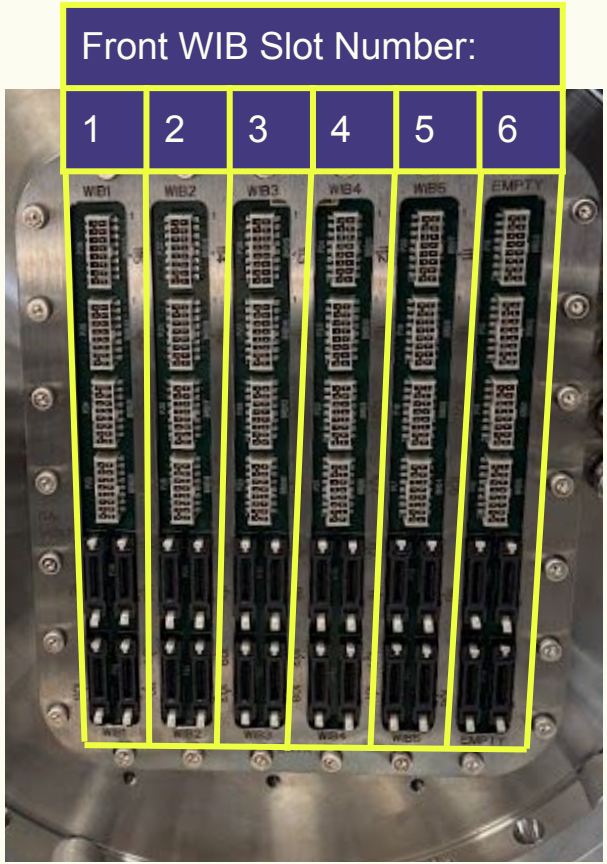
Attach PTC Cables.
Screw in lower connector.



PTC Installed

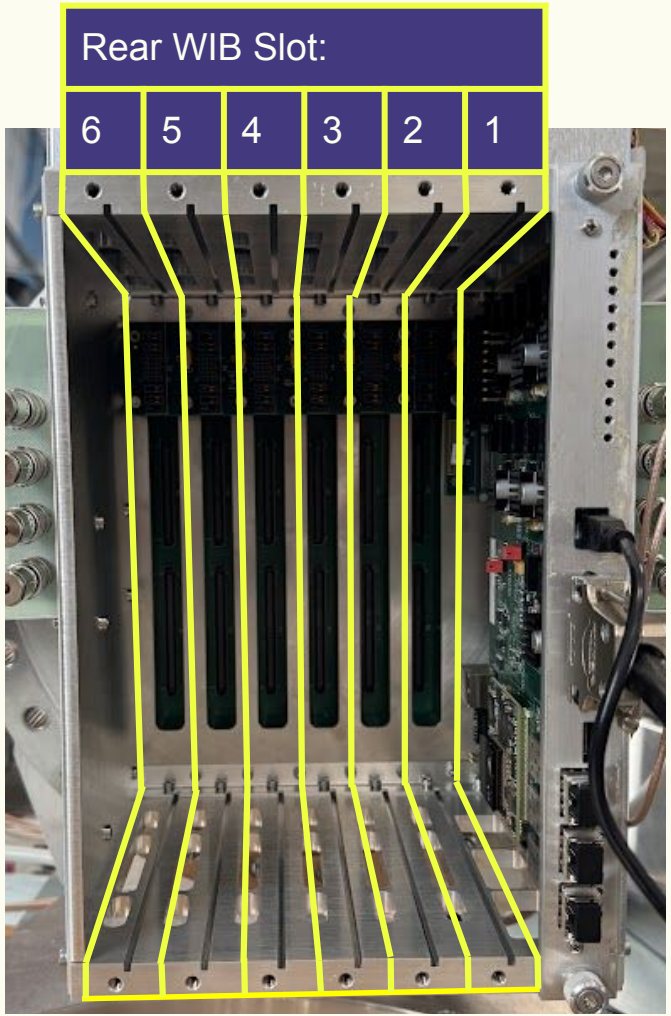
# WIB/FEMB Test

## WIB Slot Mapping




On both the front and the rear of the WIEC, the WIB slot closest to the fan side is slot 1.

Testing will go in order from slot 1 to slot 6.

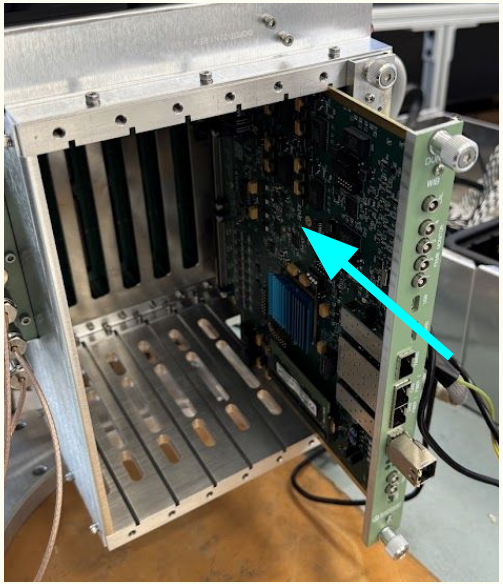


# WIB/FEMB Test

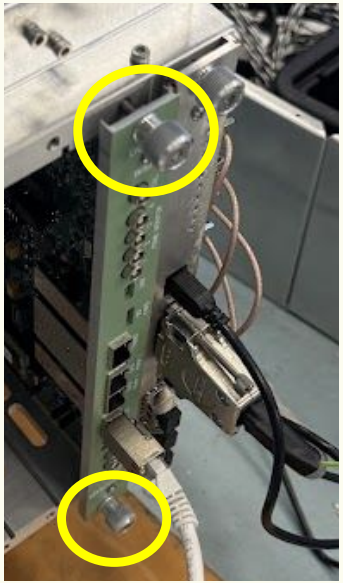
## Install WIB in Slot 1

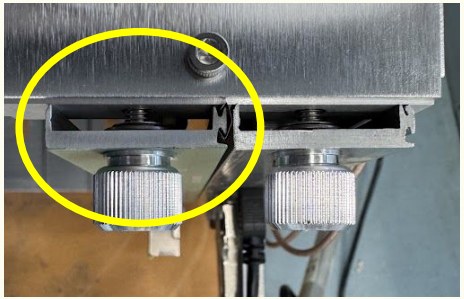


WIB



Slide the WIB into the slot closest to the PTC until you can screw in the top and bottom screws.





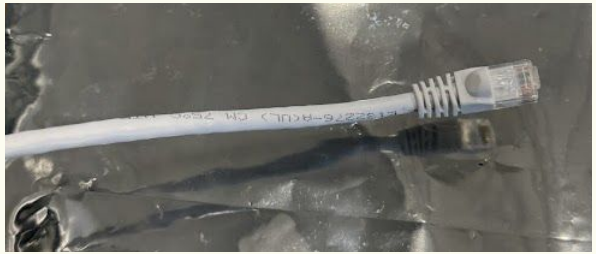
Screw in the screws until it becomes difficult to screw them further.

Press the board further into the slot.

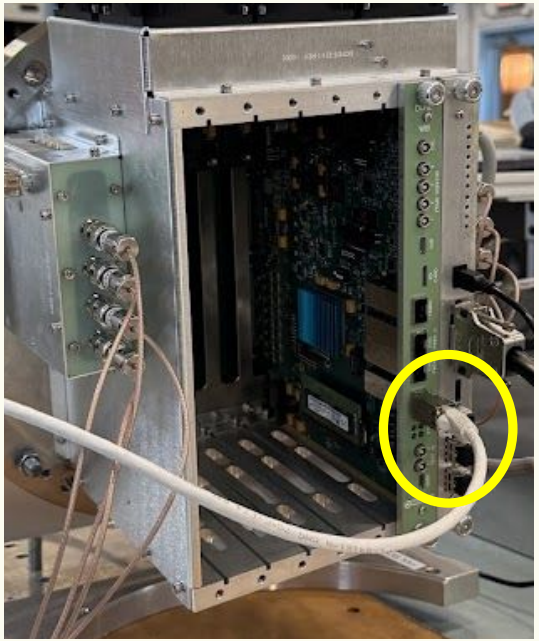
Alternate screwing in and pressing the WIB until the board is flush with the crate.

# WIB/FEMB Test

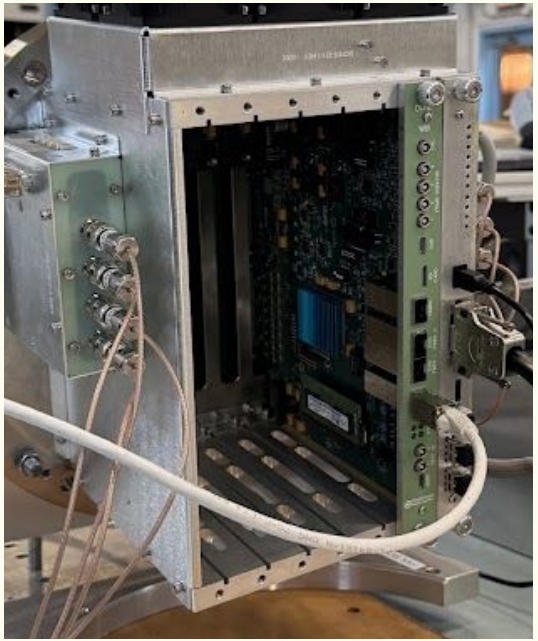
## Install Ethernet cable on WIB



Ethernet Cable



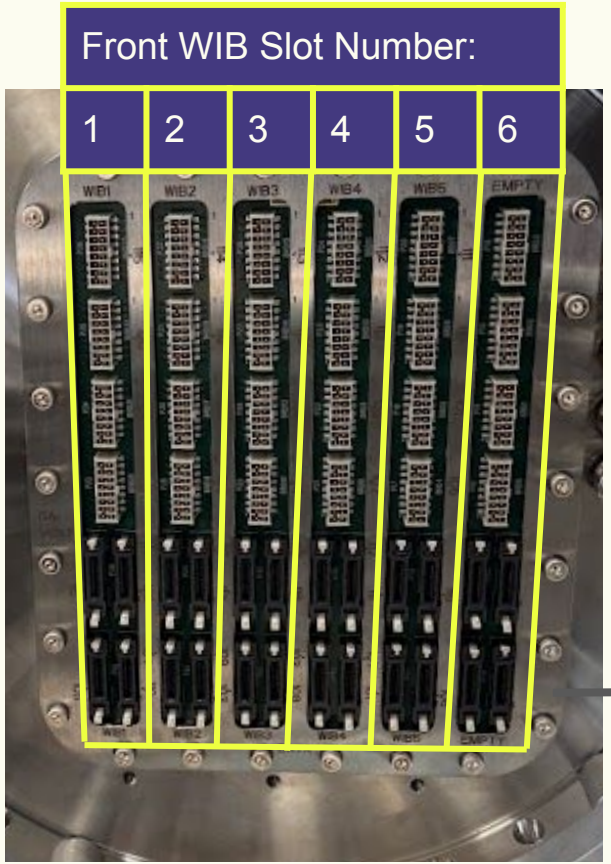
Plug the ethernet cable into the WIB.



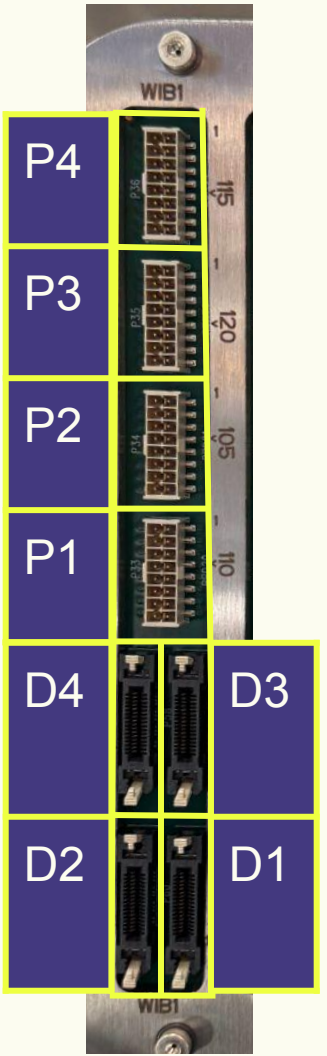
WIB installed in slot 1.

# WIB/FEMB Test

## FEMB Connector Mapping

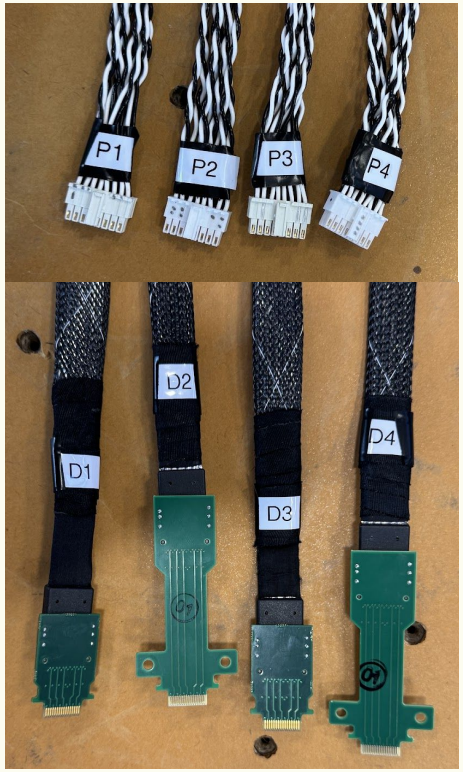



- Each WIB slot has 8 connectors on the front of the WIEC.
- The same FEMB cables will be used for each WIB slot.
- Make sure you connect the FEMB cables to the correct WIB slot for the test.

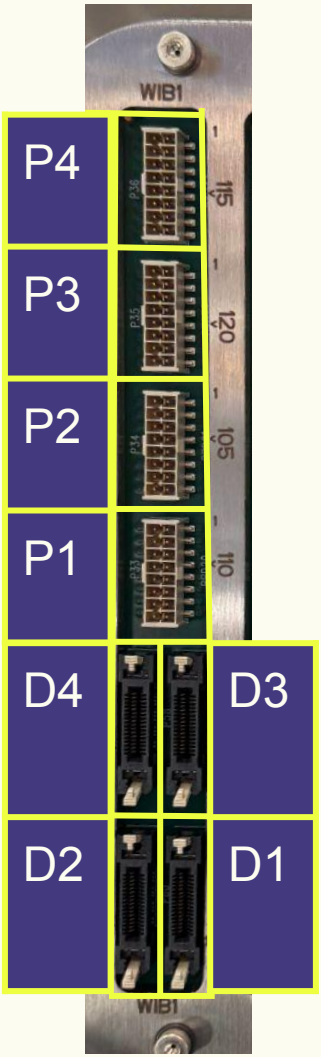


# WIB/FEMB Test

## FEMB Install




FEMB Cables



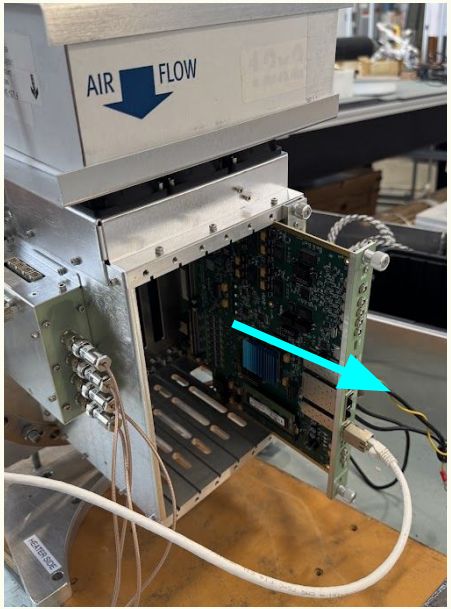

- Locate WIB test slot 1 on the front of the WIEC.
- Open the tabs on the **D** connectors.
- The labels on the cables must face towards the HEATER SIDE.
- Install the **P** cables; make sure they click into place.
- Install the **D** cables; close the tabs on those connectors.



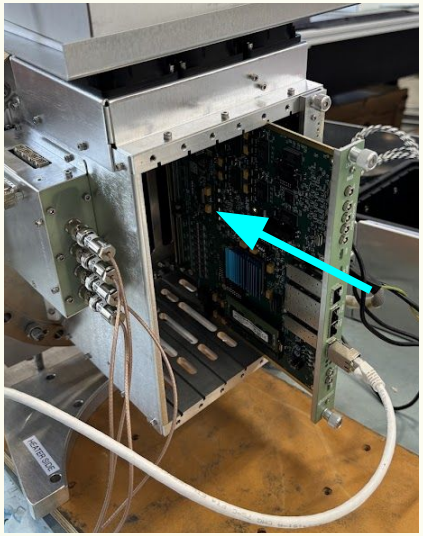

FEMB cables installed in front WIB slot 1.

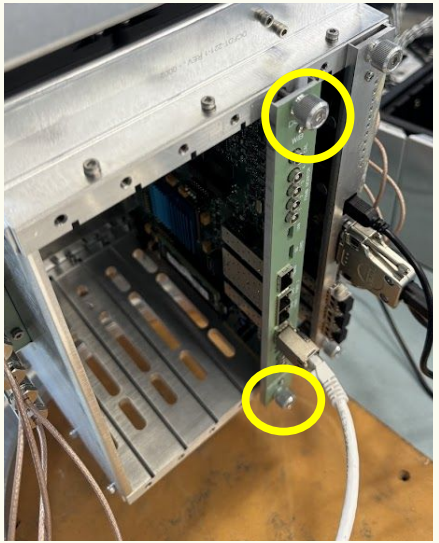
# WIB/FEMB Test

## Move WIB to next slot




Unscrew screws, pull WIB out of slot. Ethernet cable can remain connected.



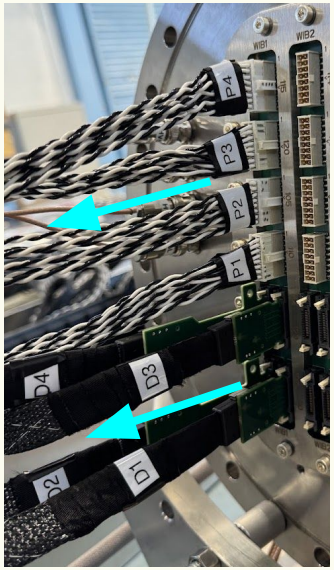




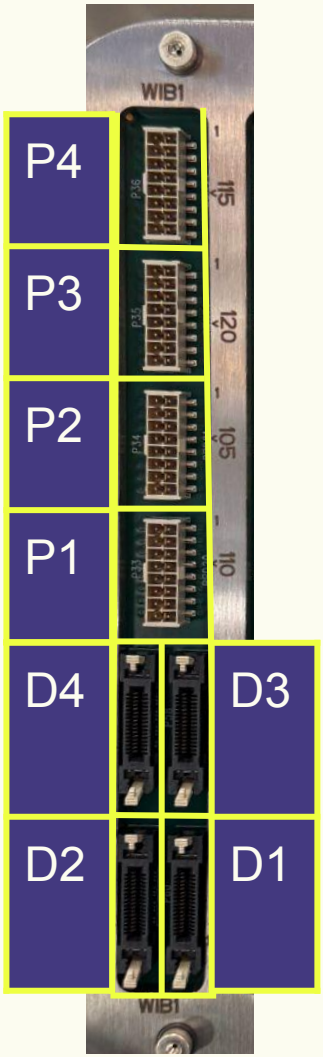
- Slide the WIB into the next slot towards the HEATER side.
- Screw in the screws until it becomes difficult to screw them further.
- Press the board further into the slot.
- Alternate screwing in and pressing the WIB until the board is flush with the crate.

# WIB/FEMB Test

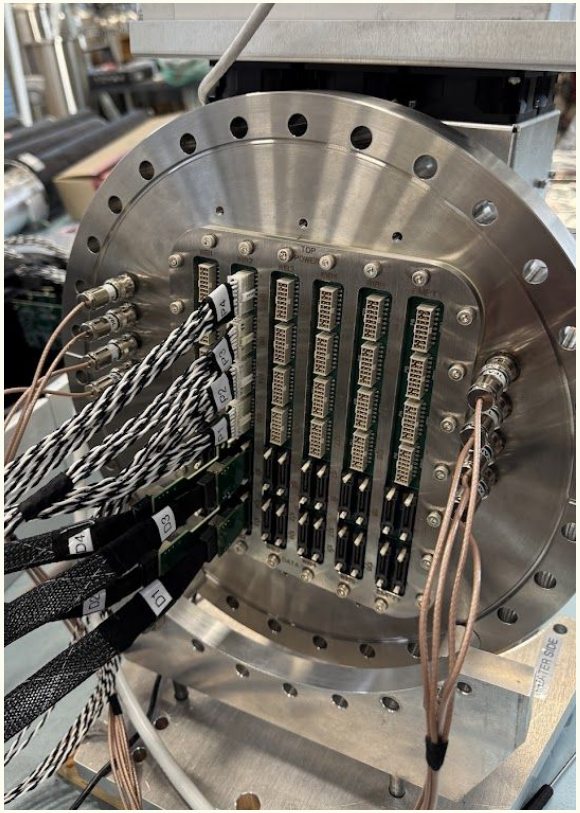
## Move FEMB Cables




Open tabs around the **D** FEMB cables. Remove all FEMB cables.



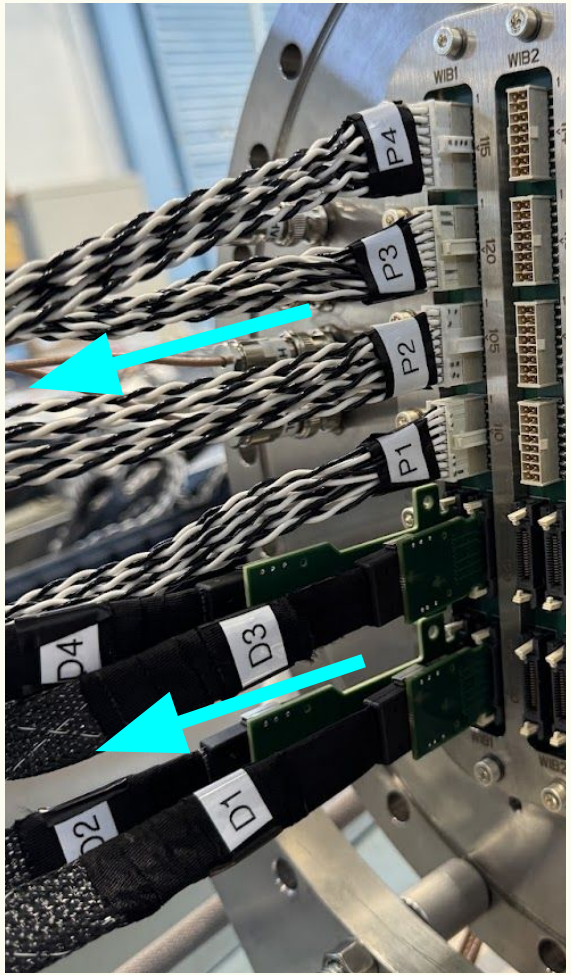

- Locate the WIB test slot on the front of the WIEC.
- Open the tabs on the **D** connectors.
- The labels on the cables must face towards the HEATER SIDE.
- Install the **P** cables; make sure they click into place.
- Install the **D** cables; close the tabs on those connectors.



FEMB Cables installed.

# Disassembly

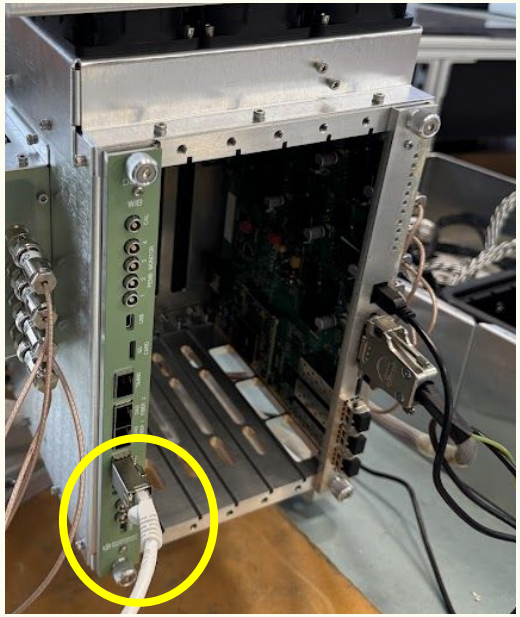
## Remove FEMB Cables

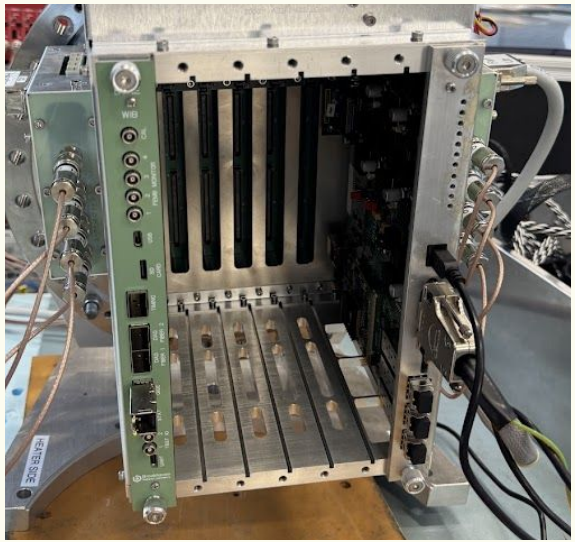



- Open tabs around the **D** FEMB connectors.
- Remove all FEMB cables from their slots.
- Return FEMB cables to their proper storage location.

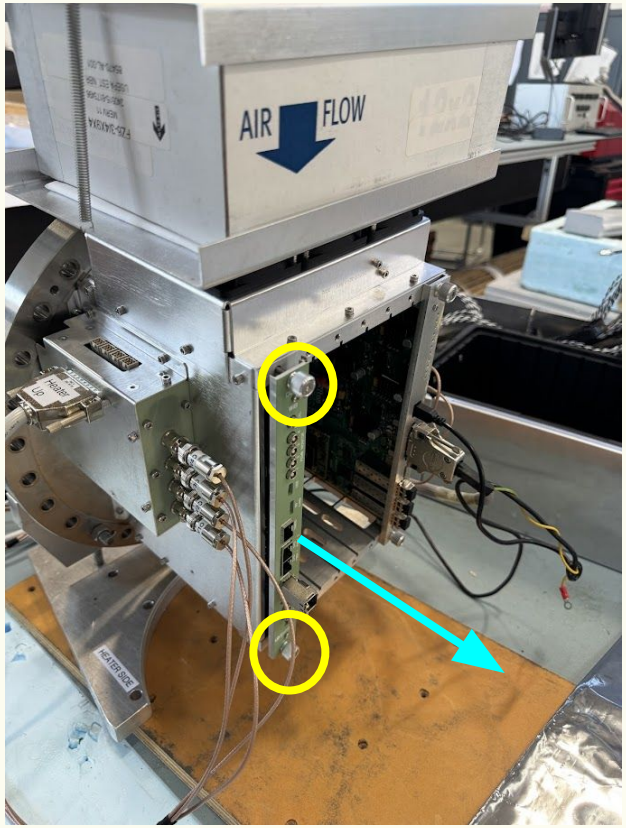
# Disassembly

## Remove WIB





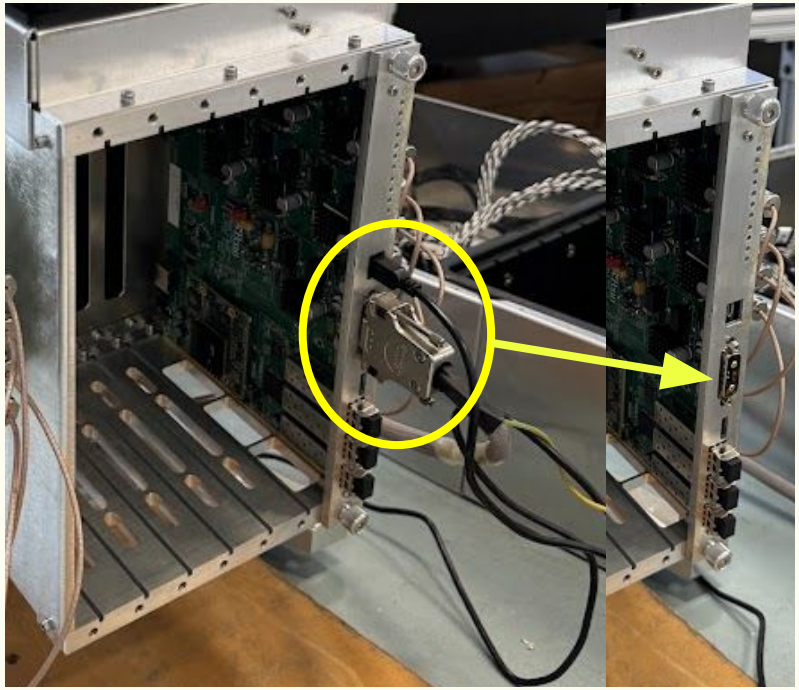
Remove Ethernet cable from the WIB.




Unscrew WIB, pull out of slot.
Return WIB to its storage site.

# Disassembly

## Remove PTC



Unscrew and remove cables from the PTC.

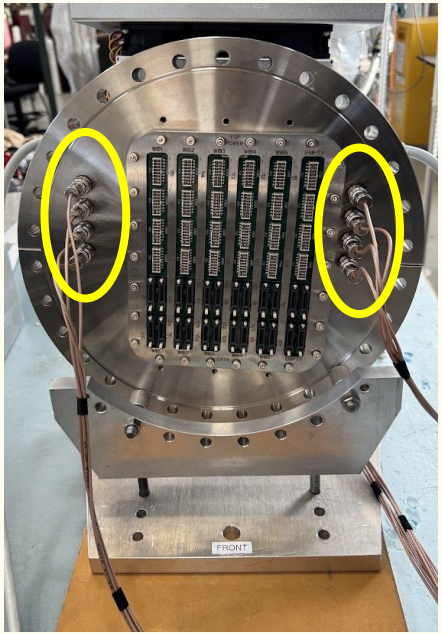



Unscrew PTC, pull out of slot.
Return PTC to its storage site.

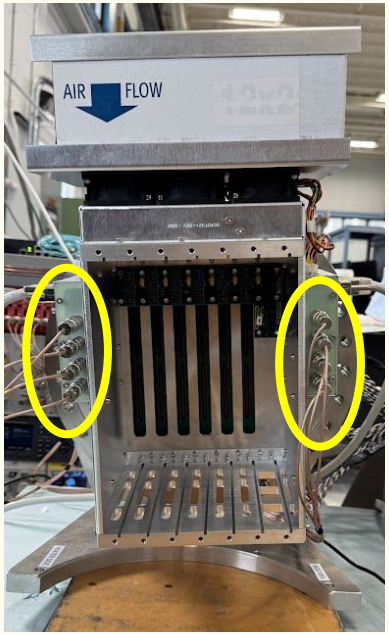
# Disassembly

## Remove Cabling

Remove SHV Cables.

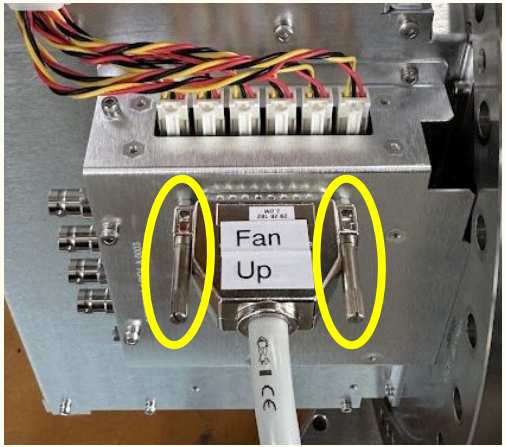



x8 on front

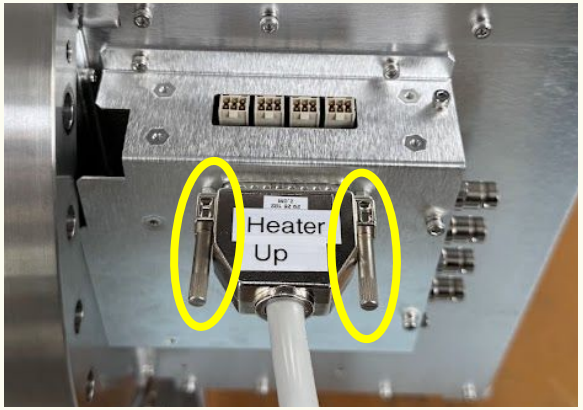



x8 on rear

Fan Side




Heater Side




Unscrew the circles screws.
Remove the Fan and Heater Cables.

WIEC QC

# PASS

WIEC QC

# FAIL

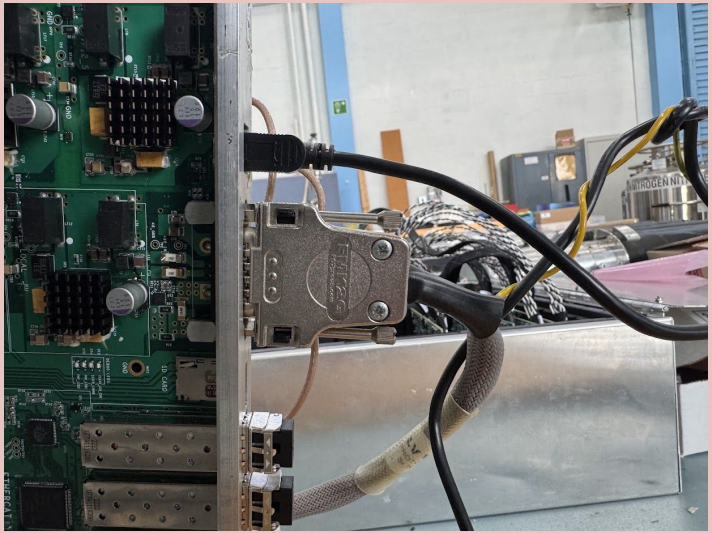
Check the popup report for the cause.

# Error

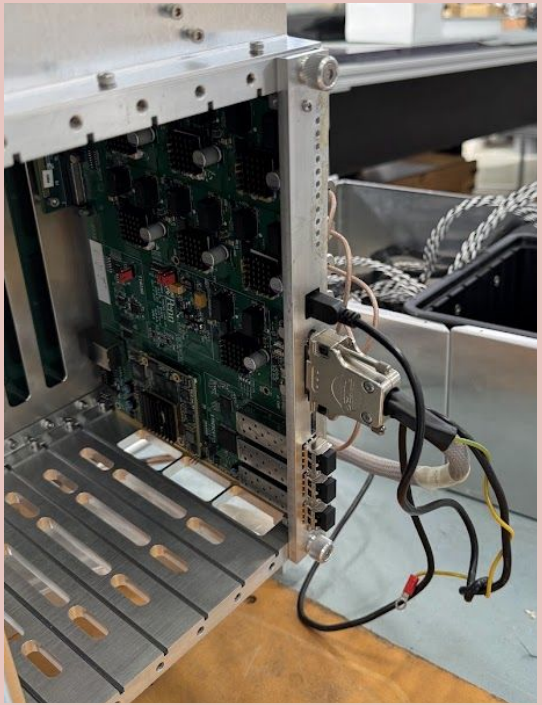
## PTC Startup Failed



Is the PTC screwed in fully and flush with the crate?



Are the PTC cables fully seated in their connectors? Are the screws all the way in?



This is what proper PTC installation looks like.

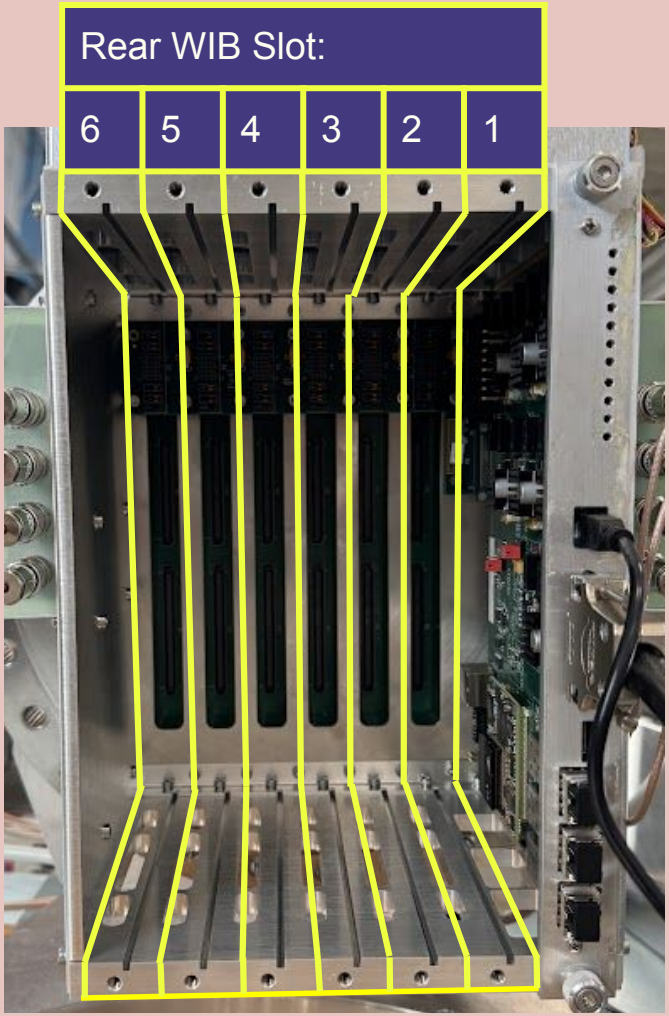
# Error

## WIB Checkout Failed

- Is the WIB installed in the same slot as the test number indicated below?
- Is the WIB pushed all the way into the slot and flush with the crate?
- Is the WIB's ethernet cable installed fully?



WIB flush with crate with ethernet installed.



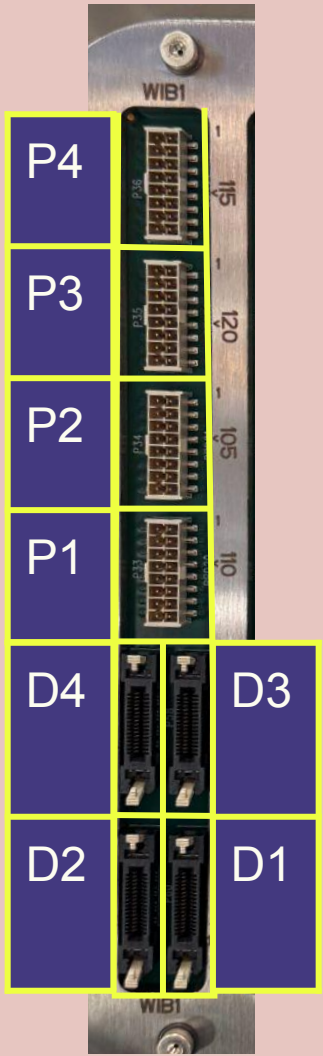
# Error

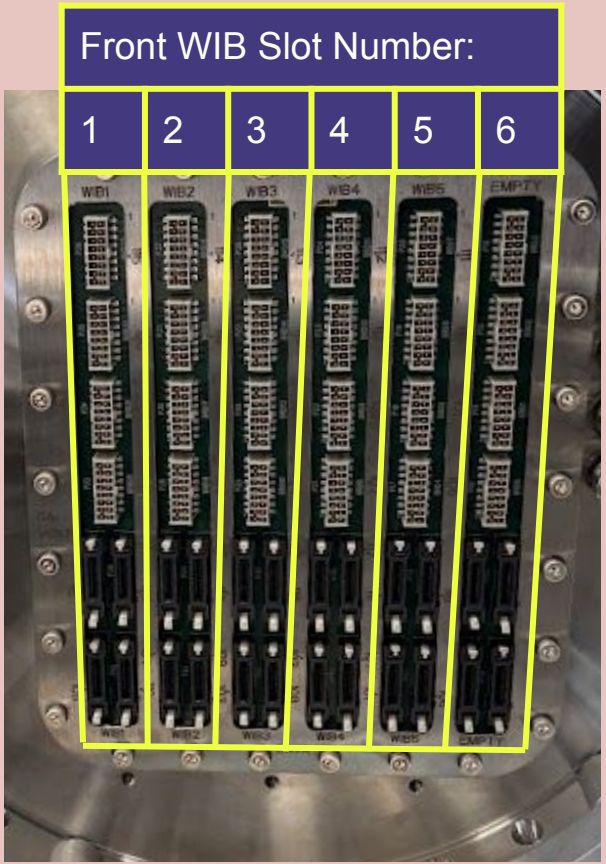
## FEMB Checkout Failed




- Are the labels on the FEMB cables all facing the heater side?

- Are the FEMB cables in the correct connectors?
- Are the cables all fully inserted into their connectors?




- Are the FEMB cables plugged into the correct WIB slot?